Supplementary Data

`lifelines-hc`: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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This document accompanies the Application Note *lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards*. It provides (i) the full analysis workflow and scripts used to reproduce the case studies, and (ii) complete per-method test statistics and p-values for all four datasets analysed in the main text.

1 Analysis scripts and data

All scripts live in the `AppNote/` directory of the `lifelines-hc` repository (`https://github.com/alonkipnis/lifelines-hc/tree/main/AppNote`). They require Python ≥ 3.10, `lifelines-hc`, and `lifelines` ≥ 0.29.

Workflow

1. **Download trial IPD.** `01_get_kmdata.py` exports reconstructed individual patient data (IPD) from the `kmdata` R package [Guyot et al., 2012] for CheckMate 057 [Borghaei et al., 2015], COMET-1 [Smith et al., 2016], and AZURE [Coleman et al., 2011, 2014]. CSL liver-cirrhosis IPD are downloaded directly from the CRAN `timereg` package and the SurvSet repository [Martinussen and Scheike, 2006, Drysdale, 2022] (see `run_csl.py`).

2. **Run per-dataset analyses.** Each `run_*.py` script loads one dataset, calls `run_all_tests()` in `utils.py`, writes a results CSV to `results/`, and saves diagnostic figures to `figs/`.

3. **Build the composite figure.** `make_figure.py` assembles Figure 1 of the main text from the four domain scripts.

Script index

Reproduction commands

```
cd AppNote
python 01_get_kmdata.py --trials Checkmate057_1C COMET1_2A AZURE_2A
python run_immuno_oncology.py
python run_comet.py
python run_azure.py
python run_csl.py          # requires SurvSet or AppNote/data/CSL.csv
python make_figure.py
```

Each `run_*.py` script writes `results/<domain>_test_results.csv` containing one row per competing method (statistic and permutation- or asymptotic-based p-value).

Table 1: Analysis scripts for the four case studies.

Script	Dataset	Role
`01_get_kmdata.py`	CheckMate 057, COMET-1, AZURE	Download IPD via `kmdata` (Guyot algorithm).
`run_immuno_oncology.py`	CheckMate 057 PFS ($n = 582$)	Nivolumab vs. docetaxel; crossing curves.
`run_comet.py`	COMET-1 OS ($n = 1028$)	Cabozantinib vs. prednisone; transient early benefit.
`run_azure.py`	AZURE DFS ($n = 3359$)	Zoledronic acid vs. control; menopause-dependent effect.
`run_csl.py`	CSL OS ($n = 446$)	Prednisone vs. placebo; public raw IPD, multi-window effect.
`utils.py`	—	Shared `run_all_tests()`, plotting, and table utilities.
`make_figure.py`	All four	Composite Figure 1 panels (A–H).

2 Data sources

Table 2 lists the original clinical studies, endpoints, and how each dataset was obtained. For CheckMate 057, COMET-1, and AZURE, only published Kaplan–Meier curves were available; IPD were reverse-engineered with the algorithm of Guyot et al. [2012] and distributed in the `kmdata` package (https://github.com/raredd/kmdata). The CSL trial is the only case study with genuinely public raw IPD.

Table 2: Original studies and data acquisition for the four case studies.

Trial	Endpoint	n	Source and acquisition
CheckMate 057	PFS	582	Borghaei et al. [2015]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (`kmdata`).
COMET-1	OS	1028	Smith et al. [2016]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (`kmdata`).
AZURE	DFS	3359	Coleman et al. [2011, 2014]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (`kmdata`).
CSL	OS	446	Copenhagen liver-cirrhosis trial (1962–1969); raw IPD from the CRAN `timereg` package [Martinussen and Scheike, 2006] and SurvSet [Drysdale, 2022].

Reconstructed IPD files are written to `AppNote/data/` as `Checkmate057_1C.csv`, `COMET1_2A.csv`, and `AZURE_2A.csv` by `01_get_kmdata.py`. CSL data are written to `AppNote/data/CSL.csv` by `run_csl.py`.

3 Shared analysis settings

Unless noted otherwise, all HC-based tests (HCHG, Fisher combination, MinP) use:

- equal-width time pooling with `n_intervals_to_pool` as listed per dataset below;
- two-sided interval hypergeometric p-values (`alternative='both'`);
- HC fraction $\gamma = 0.2$;
- label-permutation p-values with the random seed fixed at 42.

The log-rank test and four weighted log-rank variants (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington with $p = q = 1$) use the standard `lifelines.statistics.logrank_test` implementation with asymptotic p-values.

4 Per-method results tables

Tables 3–6 report the full comparison for each dataset. Asterisks mark $p < 0.05$. Permutation-based p-values for HCHG, Fisher, and MinP are discrete (multiples of $1/(n_{\text{perms}} + 1)$ with $n_{\text{perms}} = 2000$) and are rounded to three decimals; "< 0.001" denotes the smallest attainable value, 1/2001.

CheckMate 057 — progression-free survival ($n = 582$)

Original trial: Borghaei et al. [2015]. IPD reconstructed from published Kaplan–Meier curves [Guyot et al., 2012]. Control: docetaxel ($n = 290$); treatment: nivolumab ($n = 292$). HC settings: $K = 60$ intervals, $n_{\text{perms}} = 2000$.

Table 3: CheckMate 057 PFS: all competing methods.

Method	Statistic	p-value
Log-rank	0.868	0.351
Gehan-Wilcoxon	4.169	0.041*
Tarone-Ware	0.844	0.358
Peto-Prentice	3.869	0.049*
Fleming-Harrington (1,1)	1.292	0.256
Higher Criticism (HC)	1.695	0.001*
Fisher combination	112.561	< 0.001*
MinP (Bonferroni)	7.112	0.007*

COMET-1 — overall survival ($n = 1028$)

Original trial: Smith et al. [2016]. IPD reconstructed from published Kaplan–Meier curves [Guyot et al., 2012]. Control: prednisone ($n = 346$); treatment: cabozantinib ($n = 682$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

AZURE — disease-free survival ($n = 3359$)

Original trial: Coleman et al. [2011, 2014]. IPD reconstructed from published Kaplan–Meier curves [Guyot et al., 2012]. Control: standard adjuvant therapy ($n = 1678$); treatment: zoledronic acid ($n = 1681$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

Table 4: COMET-1 OS: all competing methods.

Method	Statistic	p-value
Log-rank	1.259	0.262
Gehan-Wilcoxon	1.693	0.193
Tarone-Ware	1.570	0.210
Peto-Prentice	1.602	0.206
Fleming-Harrington (1,1)	0.648	0.421
Higher Criticism (HC)	1.179	0.014*
Fisher combination	128.932	0.028*
MinP (Bonferroni)	4.858	0.242

Table 5: AZURE DFS: all competing methods.

Method	Statistic	p-value
Log-rank	1.054	0.305
Gehan-Wilcoxon	0.652	0.419
Tarone-Ware	0.691	0.406
Peto-Prentice	1.145	0.284
Fleming-Harrington (1,1)	0.277	0.598
Higher Criticism (HC)	1.513	0.005*
Fisher combination	157.060	0.004*
MinP (Bonferroni)	9.168	0.002*

CSL — overall survival ($n = 446$)

Original trial: Copenhagen Study group for Liver diseases (1962–1969); public IPD distributed via Martinussen and Scheike [2006] and Drysdale [2022]. Control: placebo ($n = 220$); treatment: prednisone ($n = 226$). Follow-up times are in years. HC settings: $K = 50$ intervals, $n_{\text{perms}} = 2000$.

Table 6: CSL liver-cirrhosis OS: all competing methods.

Method	Statistic	p-value
Log-rank	0.757	0.384
Gehan-Wilcoxon	0.439	0.507
Tarone-Ware	0.774	0.379
Peto-Prentice	0.574	0.449
Fleming-Harrington (1,1)	2.234	0.135
Higher Criticism (HC)	1.257	0.002*
Fisher combination	77.413	0.034*
MinP (Bonferroni)	5.291	0.057

5 Cross-dataset summary

Table 7 condenses the log-rank and HCHG p-values reported in the main text. Full statistics for all eight methods appear in Section 4.

Table 7: Summary of log-rank and HCHG p-values across the four case studies (full tables in Section 4).

Dataset	Endpoint	n	Log-rank p	HCHG p
CheckMate 057	PFS	582	0.351	0.001
COMET-1	OS	1028	0.262	0.014
AZURE	DFS	3359	0.305	0.005
CSL	OS	446	0.384	0.002

Across all four datasets, HCHG achieves $p \leq 0.014$ while the log-rank test is non-significant throughout ($p \geq 0.26$). Among the weighted log-rank alternatives, only CheckMate 057 Gehan-Wilcoxon and Peto-Prentice reach $p < 0.05$ (at 0.041 and 0.049 respectively); every other weighted test returns $p \geq 0.13$.

6 Machine-readable results

The CSV files below contain the same values as Tables 3–6 (columns `method`, `statistic`, `p_value`):

- `AppNote/results/immuno_test_results.csv`
- `AppNote/results/comet_test_results.csv`
- `AppNote/results/azure_test_results.csv`
- `AppNote/results/csl_test_results.csv`

References

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